

Project description: Deployment of a SIEM/XDR Solution with Wazuh

The objective of this project was to establish a centralized security monitoring environment to detect threats and manage compliance on a Windows endpoint. To achieve this, I deployed a **Wazuh Manager** on a Linux server and configured a remote agent to collect and analyze security events in real-time.

Step 1: Infrastructure Setup (Server-side)

First, I deployed an **Ubuntu Server** instance using Oracle VirtualBox, configuring the network adapter in **Bridged Mode** to ensure local connectivity. I then executed the Wazuh installation script to stand up the Indexer, Server, and Dashboard.

```
File Machine View Input Devices Help
23/01/2026 02:58:11 INFO: Generating Wazuh dashboard certificate
23/01/2026 02:58:12 INFO: Created wazuh-install-files.tar. It
23/01/2026 02:58:13 INFO: --- Wazuh indexer ---
23/01/2026 02:58:13 INFO: Starting Wazuh indexer installation
23/01/2026 03:01:37 INFO: Wazuh indexer installation finished
23/01/2026 03:01:37 INFO: Wazuh indexer post-install configuration
23/01/2026 03:01:37 INFO: Starting service wazuh-indexer.
23/01/2026 03:02:19 INFO: wazuh-indexer service started.
23/01/2026 03:02:19 INFO: Initializing Wazuh indexer cluster
23/01/2026 03:02:34 INFO: Wazuh indexer cluster security configuration
23/01/2026 03:02:34 INFO: Wazuh indexer cluster initialized.
23/01/2026 03:02:34 INFO: --- Wazuh server ---
23/01/2026 03:02:34 INFO: Starting the Wazuh manager installation
23/01/2026 03:05:28 INFO: Wazuh manager installation finished
23/01/2026 03:05:28 INFO: Wazuh manager vulnerability detection
23/01/2026 03:05:28 INFO: Starting service wazuh-manager.
23/01/2026 03:05:51 INFO: wazuh-manager service started.
23/01/2026 03:05:51 INFO: Starting Filebeat installation.
23/01/2026 03:06:13 INFO: Filebeat installation finished.
23/01/2026 03:06:14 INFO: Filebeat post-install configuration
23/01/2026 03:06:14 INFO: Starting service filebeat.
23/01/2026 03:06:18 INFO: filebeat service started.
23/01/2026 03:06:18 INFO: --- Wazuh dashboard ---
23/01/2026 03:06:18 INFO: Starting Wazuh dashboard installation
23/01/2026 03:09:36 INFO: Wazuh dashboard installation finished
23/01/2026 03:09:36 INFO: Wazuh dashboard post-install configuration
23/01/2026 03:09:36 INFO: Starting service wazuh-dashboard.
23/01/2026 03:09:37 INFO: wazuh-dashboard service started.
23/01/2026 03:09:41 INFO: Updating the internal users.
23/01/2026 03:09:55 INFO: A backup of the internal users has been
23/01/2026 03:10:24 INFO: The filebeat.yml file has been updated
23/01/2026 03:11:14 INFO: Initializing Wazuh dashboard web application
23/01/2026 03:11:15 INFO: Wazuh dashboard web application not
23/01/2026 03:11:30 INFO: Wazuh dashboard web application not
23/01/2026 03:11:45 INFO: Wazuh dashboard web application initialized
23/01/2026 03:11:45 INFO: --- Summary ---
23/01/2026 03:11:45 INFO: You can access the web interface at
User:
Password:
```

IP 192.168.X.XXX

This setup provides the central "brain" of the SIEM, where all logs will be processed and visualized through the web interface.

Step 2: Agent Deployment and Troubleshooting

To monitor my host machine, I installed the **Wazuh Agent for Windows**. Initially, the agent failed to report to the manager due to a configuration mismatch.

I used **Windows PowerShell (Admin)** to manually register the agent: & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m **192.168.X.XXX**

Administrator: Windows PowerShell

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Windows PowerShell

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PS C:\WINDOWS\system32>

```
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m [REDACTED]
2026/01/23 16:07:35 agent-auth: INFO: Started (pid: 5972).
2026/01/23 16:07:35 agent-auth: INFO: Requesting a key from server: [REDACTED]
2026/01/23 16:07:58 agent-auth: ERROR: (1208): Unable to connect to enrollment service at '[REDACTED]:1515'
PS C:\WINDOWS\system32> & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m [REDACTED]
2026/01/23 16:12:27 agent-auth: INFO: Started (pid: 12168).
2026/01/23 16:12:27 agent-auth: INFO: Requesting a key from server: [REDACTED]
2026/01/23 16:12:27 agent-auth: INFO: No authentication password provided
2026/01/23 16:12:27 agent-auth: INFO: Using agent name as [REDACTED]
2026/01/23 16:12:27 agent-auth: INFO: Waiting for server reply
2026/01/23 16:12:28 agent-auth: INFO: Valid key received
PS C:\WINDOWS\system32> net start wazuh-agent
The service name is invalid.

More help is available by typing NET HELPMSG 2185.

PS C:\WINDOWS\system32> Start-Service -Name "Wazuh"
PS C:\WINDOWS\system32> Start-Service -Name "wazuh-agent"
Start-Service : Cannot find any service with service name 'wazuh-agent'.
At line:1 char:1
+ Start-Service -Name "wazuh-agent"
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (wazuh-agent:String) [Start-Service], ServiceCommandException
+ FullyQualifiedErrorId : NoServiceFoundForGivenName,Microsoft.PowerShell.Commands.StartServiceCommand

PS C:\WINDOWS\system32> net start Wazuh
The Wazuh service is starting.
The Wazuh service could not be started.

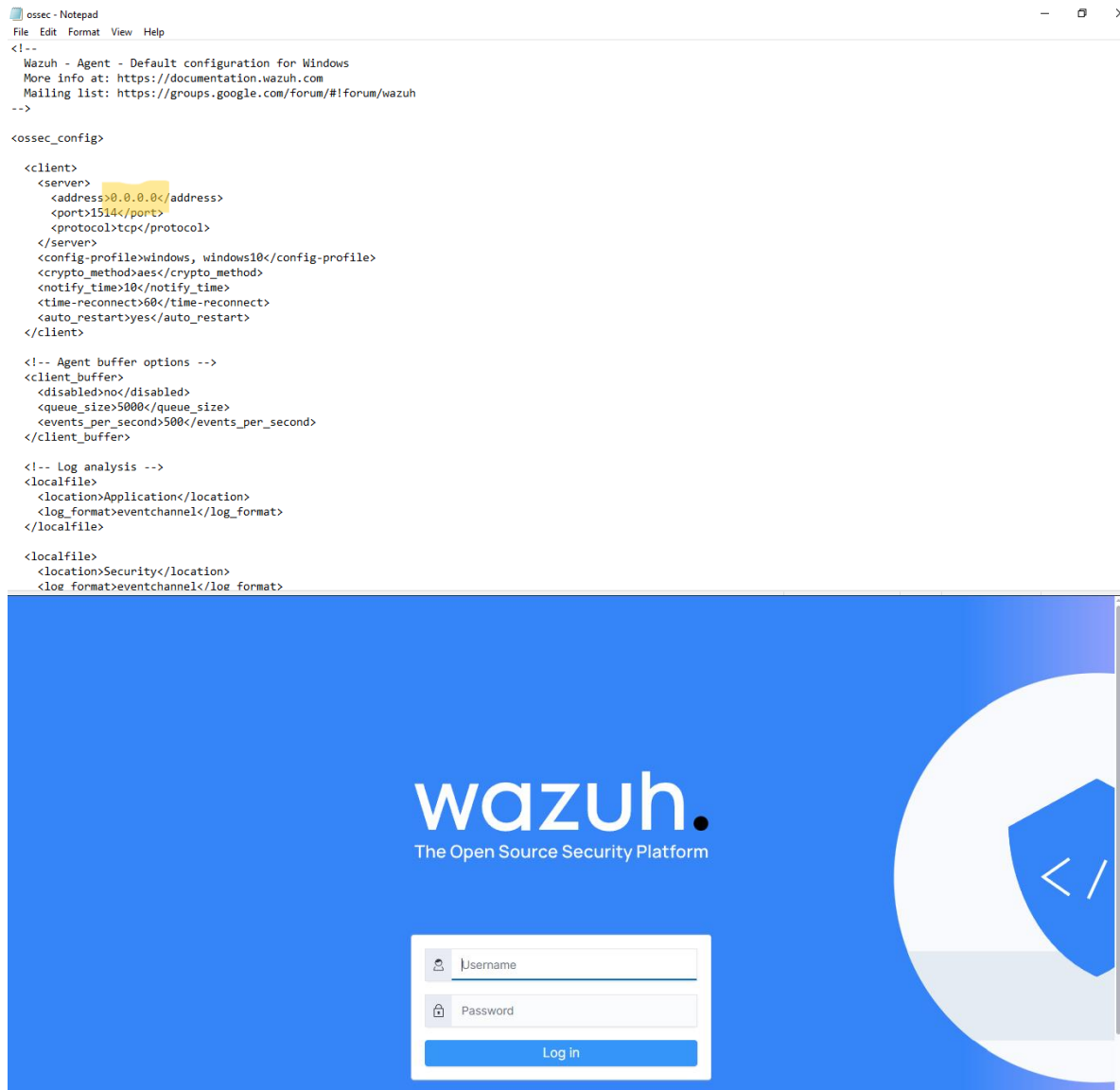
The service did not report an error.

More help is available by typing NET HELPMSG 3534.

PS C:\WINDOWS\system32>
```

Handling Technical Issues

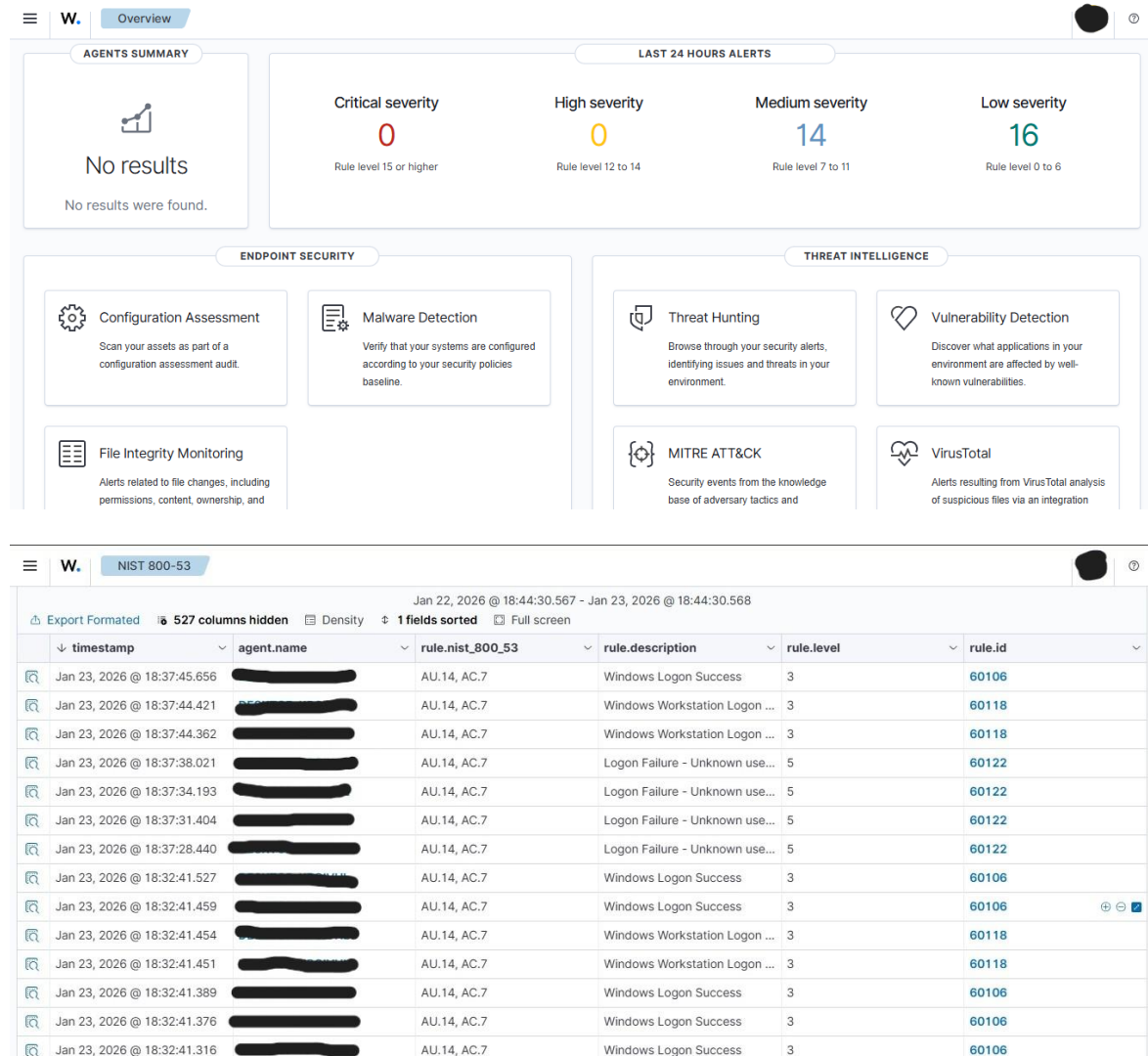
I encountered an issue where the agent service wouldn't start. Upon inspecting the ossec.conf file, I discovered the manager address was set incorrectly to 0.0.0.0. I manually edited the file using **Notepad (Admin)** to point to the correct IP:
192.168.X.XXX



Step 3: Real-time Threat Detection (Brute Force Simulation)

To validate the deployment, I simulated a **Brute Force attack** by intentionally entering incorrect credentials multiple times on the Windows login screen.

The SIEM successfully captured these events, escalating the alert severity as the attempts continued.



The system automatically categorized these events under international frameworks such as **NIST 800-53** (Controls AU.14 and AC.7) and **TSC**, providing immediate context for security auditors.